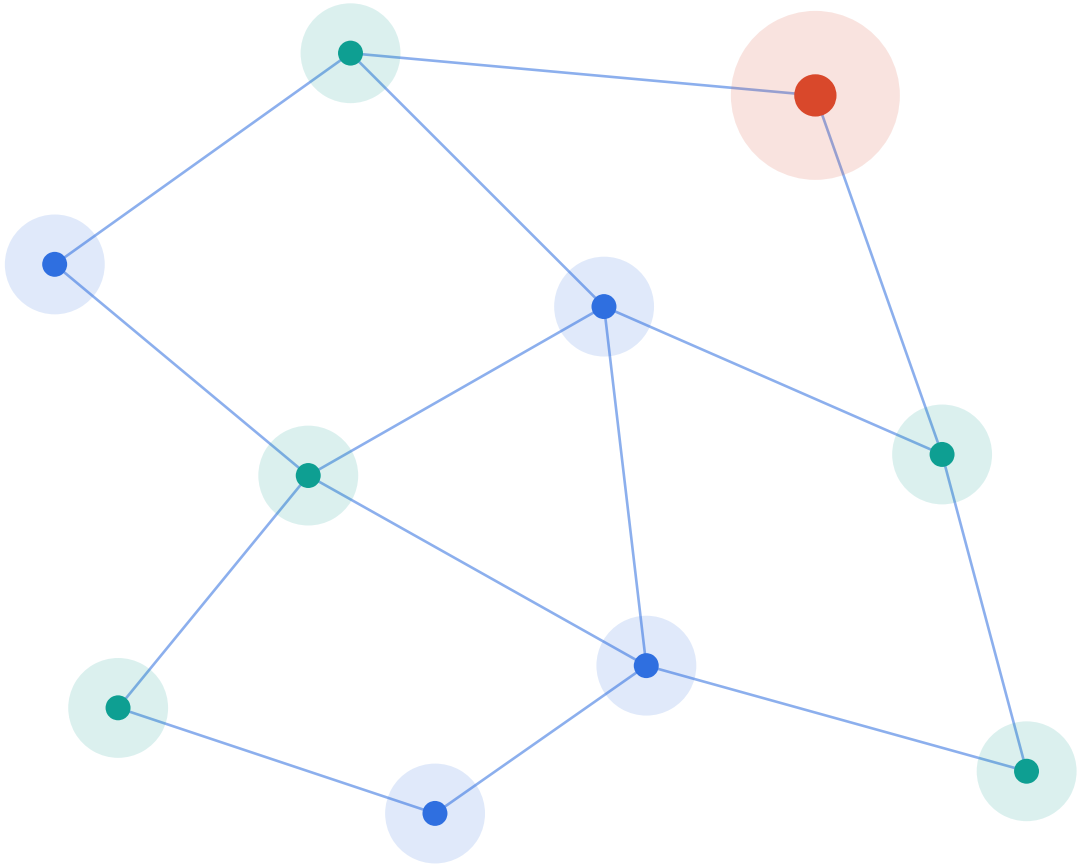
EXATHON INDIA 2026 · TRACK 01: MONEY, RISK & REGULATION

FinGuard AI

Financial Fraud Detection & Risk Intelligence Platform

Presented by Varsha Saravanan



The Problem

Normal transaction activity



Conceptual illustration, not real data

Flagged: potentially suspicious

Unusual patterns

Unusual amounts or patterns may go unnoticed

Rapid activity

Fast transaction activity can need closer examination

Alert overload

Many alerts make prioritization difficult

Slow reviews

Manual investigation workflows may consume time

Financial risk signals need to be identified, prioritized, and reviewed efficiently.

The FinGuard AI Solution

Transaction Input

Transaction information is evaluated

Heuristic Rule Checks

Implemented rules identify potentially suspicious patterns

Risk Score

Helps prioritize activity for review

Suspicious Activity Alert

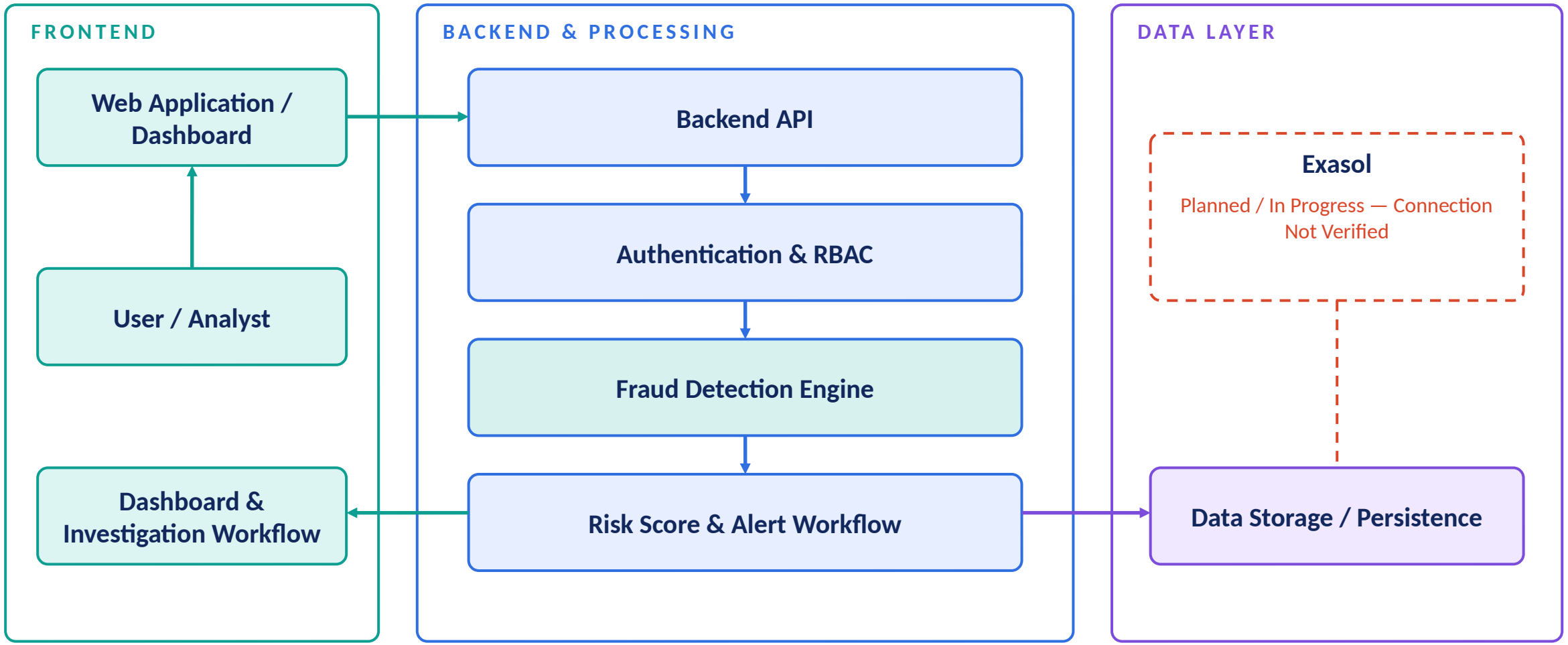
Organizes transactions that need attention

Analyst Review

Analysts review the information before deciding

A risk score does not prove that fraud has occurred. It helps prioritize what to review.

System Architecture



Solid box = prototype component Dashed = planned, not verified Arrows = data flow

Technology Stack

TypeScript web application

Typed code that keeps the project reliable

Node.js / TypeScript server

Runs the backend logic and API

Web-based user interface

Dashboard where analysts review activity

Heuristic fraud engine

Deterministic rules that produce a risk score

Authentication & RBAC

Controls who signs in and what they can do

VALIDATION NOTE

- 25 checks passed in an existing test run
- Existing production build succeeded
- These do not establish real-world fraud-detection accuracy
- Exasol connectivity not verified in the last known run

Key Advantages

Risk Prioritization

Helps organize potentially suspicious transactions for review

Structured Alert Workflow

Supports a repeatable process for examining flagged activity

Rule-Based Signals

Implemented rules can be inspected and refined

Role-Based Access

Supports permission separation where implemented

Future Expandability

A foundation for further testing and improvements

Limitations & Responsible Use

Current Limitations

- Heuristic rules can produce false positives
- Some suspicious patterns may not be detected
- Risk scoring depends on available transaction information
- Real-world detection performance has not been established by the known test results
- External integration status must be verified

Responsible Use

- Risk scores should support human review
- A flagged transaction is not automatically fraudulent
- Representative, appropriately governed data is needed for meaningful evaluation
- Rules and thresholds require testing and calibration

Future Scope

FUTURE ENHANCEMENTS, NOT COMPLETED FEATURES

1

Data Validation

Test using representative, appropriately governed, labeled transaction data

2

Performance Evaluation

Measure precision, recall, false-positive rate and relevant metrics

3

Rule Improvement

Refine thresholds and improve alert explanations

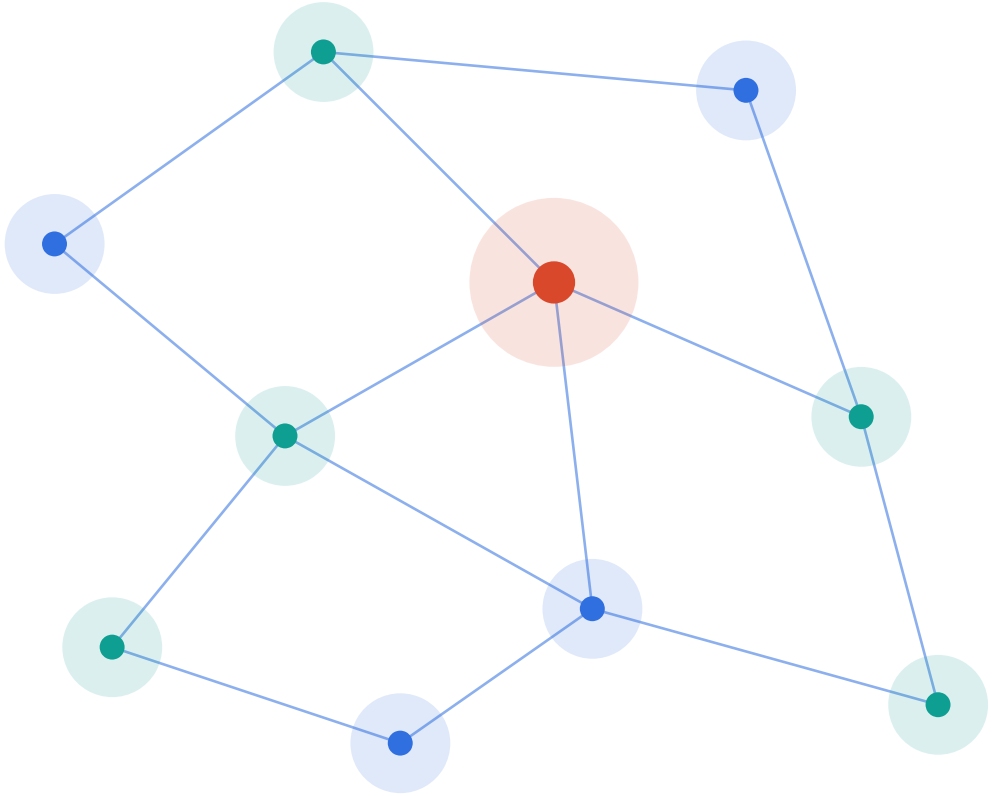
4

Integration Improvements

Complete and verify planned external integrations, including Exasol if applicable

Conclusion

Detect Patterns.
Prioritize Risk.
Support Better Investigations.



- FinGuard AI organizes transaction-risk signals
- Risk scoring can help prioritize activity for review
- Alert and investigation workflows support structured analyst review
- Further validation and integration work can strengthen the prototype